SREYA SIVAN

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SUMMARY

AI engineer and B.Tech Information Technology graduate (2026). Builds retrieval, extraction and evaluation systems on real data: LabelLens answers questions about FDA drug labels with section-level citations, extracts label sections into typed records, and measures both on every push. As an AI intern at Prayag.ai, built a business development agent in Python, automated data extraction from client emails with the OpenAI GPT-4 API and designed RAG chatbot architectures. Studying Data Science with AI at DataMites full time. Looking for AI/ML engineer and data science roles.

PROJECTS

LabelLens · Python, PostgreSQL, pgvector, sentence-transformers, pydantic, FastAPI, Ollama, pytest, GitHub Actions, Docker — 2026 | github.com/Sreyasivan/labellens

- Cited question answering over 2,000 FDA prescription labels (openFDA), chunked by label section; hybrid retrieval (bge-small vectors + Postgres full-text, reciprocal rank fusion, one chunk per drug and section). Recall@5 57/58 on a hand-written question set, up from 49/58 with naive vector search.
- Guardrails before the model: a measured distance threshold refuses all 8 off-topic questions without a model call; closed-book prompt with numbered sources; not-medical-advice footer added in code.
- Typed extraction of contraindications, interactions and warnings into pydantic models, scored against the label text: 221 items from 44 labels, 100% grounded (no item uses words absent from the label).
- Evaluation in CI on every push: each question is verified against the indexed label text, pgvector is stood up, the eval drugs ingested, and the build fails below a recall floor; 39 unit tests. Local 3B model by default so label text never leaves the machine; OpenAI or Claude via environment variables.

FAERS Serious-Outcome Classifier · Python, pandas, scikit-learn, matplotlib, pytest, GitHub Actions — 2026 | github.com/Sreyasivan/faers-serious-outcome

- Predicts death, life-threatening event or hospitalisation on 397,224 real FDA adverse-event reports (one FAERS quarter): logistic-regression baseline AUROC 0.83 vs gradient boosting 0.87 (AUPRC 0.72), both calibrated across ten probability bins; permutation importance and a model card with intended use and failure modes.
- Found two label leaks by EDA (outcome-naming reaction terms; expedited-vs-periodic report type at 48% vs 3% serious) and removed them, reporting the lower score.

Financial Analytics & Expense Tracker · MongoDB, Express.js, React.js, Node.js, Chart.js

- Analytics dashboard with spending charts, JWT authentication and a MongoDB schema designed for the reports.

EXPERIENCE

Artificial Intelligence Intern — Prayag.ai — Palakkad, Kerala (on-site) | Jun 2025 – Apr 2026

- Built an **AI-powered business development agent** in Python.
- Integrated the **OpenAI GPT-4 API** to automate data extraction from client emails.
- Designed **RAG-based chatbot architectures**.

EDUCATION

DataMites — Data Science with AI *(certification course, full time)* — Jun 2026 – Present

- Python, machine learning, advanced data science, NLP, database management, big data foundations, AI and automation.

Karpagam Institute of Technology, Coimbatore — Nov 2022 – Apr 2026

Bachelor of Technology in Information Technology — CGPA 7.83

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, Java
AI engineering	RAG, hybrid retrieval (pgvector, Postgres full-text, RRF), embeddings (sentence-transformers), structured extraction (pydantic), evaluation harnesses, OpenAI and Anthropic APIs, Ollama
Backend and web	FastAPI, REST APIs, React.js, Node.js, Express.js, JWT authentication, Chart.js
Data	PostgreSQL, MongoDB, openFDA API; learning machine learning, NLP and big data
Tools	Git, GitHub Actions, Docker, pytest, uv

CERTIFICATIONS

- Web Security: User Authentication and Access Control — LinkedIn Learning (Feb 2025)